SMOLVLA · LIBERO · 320 + 240 EPISODES

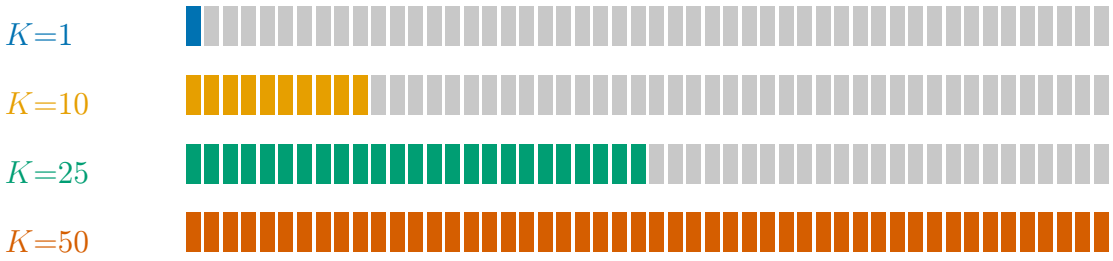
How often should a VLA policy look at the scene again?

Denis Basharin

25 October 2026

Fifty actions are predicted. Only the first K ever run.

Each strip is one call to the policy. A filled cell is an action that reaches the robot; a grey cell is an action the queue throws away. The arm does not look at the scene until the filled part has run out.



Two costs pull against each other. A short horizon spends a network call on nearly every step and, on a real arm, cannot keep up with the 20 Hz controller. A long horizon is cheap and smooth, and it keeps executing a plan the world has already invalidated.

The study changes nothing except K: one checkpoint, nine instructions, the same 80 initial states at each horizon. Replanning is synchronous, there is no ensembling across chunks, and no inference overlaps with motion.

The policy as a function, and the loop around it

At a replanning time t the network maps the current observation to a whole chunk:

$$A_t = \pi_\theta(o_t, \ell)\,, \qquad A_t \in [-1,1]^{50\times 7},$$

$$o_t = \left(I_t^{\text{agent}},\ I_t^{\text{wrist}},\ s_t\right), \qquad s_t \in \mathrm{R}^8.$$

The executed trajectory with a fixed horizon K is

$$u_{t+i} = A_t[i], \quad i = 0,\ldots,K-1, \qquad t \in \{0, K, 2K, \ldots\}.$$

Symbol	Content
$I^{\text{agent}}, I^{\text{wrist}}$	Two RGB frames, 256×256 from the simulator, rotated $180°$, padded to 512×512, scaled to $[-1,1]$.
s_t	End-effector position (3), orientation as axis-angle (3), two finger positions (2); zero-padded to 32 and standardized.
ℓ	The instruction string, tokenized, padded to 48 tokens.
$A_t[i]$	Relative translation (3), relative rotation (3), gripper (1). Positive gripper means close.
π_θ	SmolVLM2-500M backbone; an action expert on its first 16 layers; 10 Euler steps of a flow from noise of shape $(50, 32)$.

No object pose, no reward, and no previous action enters the network. Each call is a fresh read of the scene, and the only memory is the queue.

arxiv.org/abs/2506.01844

huggingface.co/HuggingFaceVLA/smolvla_libero

Nine instructions from four LIBERO suites

Short name	Instruction given to the policy	Inits	Steps	Object to pick up
libero_spatial --- the same black bowl, picked out by where it stands				
Bowl between	bowl between the plate and the ramekin, onto the plate	10	280	black bowl
Table center	bowl at the table center, onto the plate	10	280	black bowl
Bowl on stove	bowl on the stove, onto the plate	10	280	black bowl
libero_object --- one named grocery item among several, into the basket				
Alphabet soup	pick up the alphabet soup and place it in the basket	10	280	soup can
Tomato sauce	pick up the tomato sauce and place it in the basket	10	280	sauce can
Milk	pick up the milk and place it in the basket	10	280	milk carton
libero_10 --- two-stage instructions				
Soup and sauce	alphabet soup and tomato sauce, both into the basket	5	520	two cans, in turn
Stove and moka	turn on the stove and put the moka pot on it	5	520	moka pot
libero_goal --- articulated furniture				
Open drawer	open the middle drawer of the cabinet	10	300	none: pull a handle

Success is the simulator's own completion flag. The last column is the object whose pose is tracked for the failure analysis: a gripper close with no tracked object within 5 cm is a *miss*, and contact that is lost before the gripper opens is a *slip*. The drawer has no such object, so those two counts are always zero there.

Two of the scenes, at the first frame and at the frame the simulator calls done

Both runs are initial state 0 with a new plan on every step ($K{=}1$).



step 0



step 137

Alphabet soup into the basket.
The can is one of several groceries on the table. The policy must pick that one, lift it over the rim, and release. 138 control steps, 138 network calls.



step 0



step 75

Bowl between plate and ramekin.
Three black bowls are on the table; the instruction names one by position. The target is the plate. 76 control steps, 76 network calls.

Success at every horizon, and who beats whom on the same starts

	$K{=}1$	$K{=}10$	$K{=}25$	$K{=}50$
Bowl between	4/10	6/10	7/10	7/10
Table center	6/10	5/10	5/10	3/10
Bowl on stove	10/10	8/10	6/10	8/10
Alphabet soup	8/10	7/10	9/10	3/10
Tomato sauce	5/10	5/10	6/10	2/10
Milk	8/10	9/10	7/10	5/10
Soup and sauce	0/5	2/5	2/5	1/5
Stove and moka	2/5	3/5	4/5	4/5
Open drawer	8/10	7/10	8/10	4/10
Pooled, 80 starts	**51**	**52**	**54**	**37**
Network calls per episode	214	21	8.5	5.1
Control steps per episode	214	208	201	232

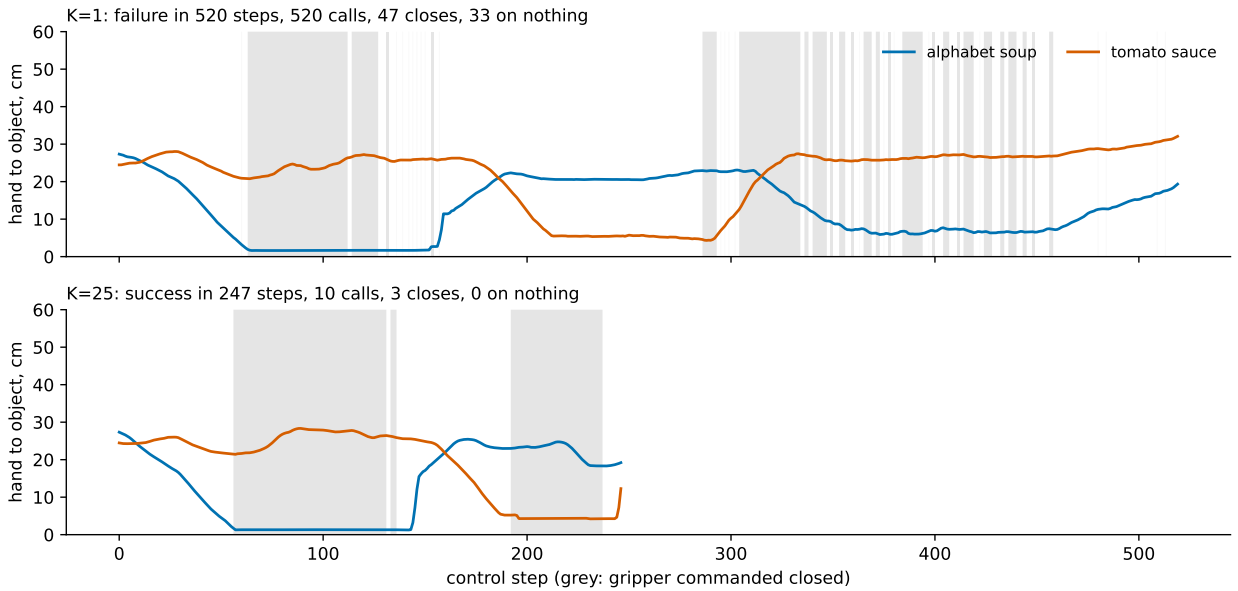
Paired on the same initial state

Pair	A wins	B wins
A = $K{=}1$, B = $K{=}50$	28	14
A = $K{=}10$, B = $K{=}50$	25	10
A = $K{=}25$, B = $K{=}50$	26	9
A = $K{=}10$, B = $K{=}25$	15	17

A ``win'' is a start that one horizon finishes and the other does not. The three shorter horizons lose to $K{=}50$ on 9 to 14 starts each, so the full chunk is not uniformly worse: it is worse on average and different in kind. Between $K{=}10$ and $K{=}25$ the wins are nearly even, 15 against 17: from 1 to 25 the pooled score moves by three starts, from 25 to 50 it drops by seventeen.

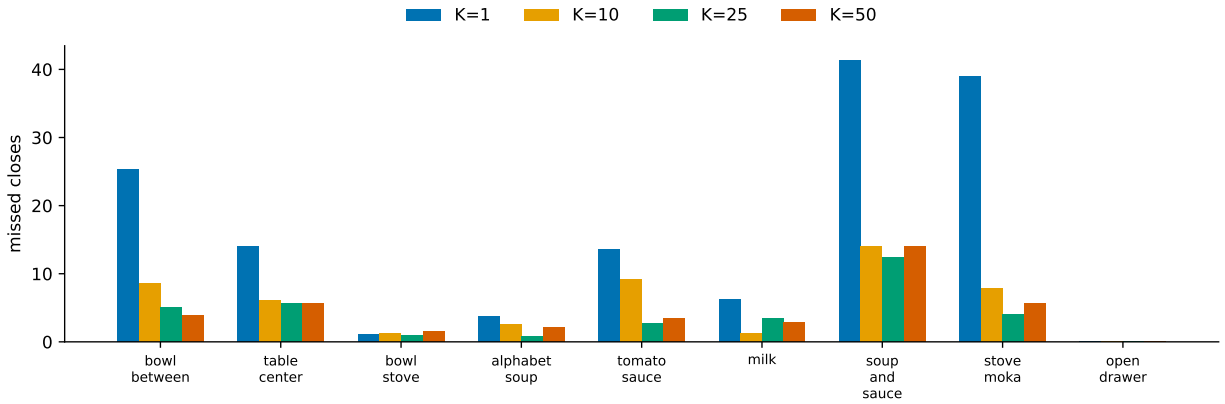
Two objects, one start: replanning every step fails

Soup and sauce, start 0. Hand-to-can distance at every step, $K{=}1$ above and $K{=}25$ below; grey bands are steps with the gripper commanded closed.



Up to step 150 the runs are identical: reach the soup, close, lift, release it in the basket; both then head for the sauce. $K{=}25$ arrives at 4 cm near step 190, closes once, lifts, and finishes at step 247. $K{=}1$ arrives at 5 cm near step 220 and hovers there for 70 steps with the gripper open: a fresh one-step plan every 50 ms never commits to the close. Near step 300 it abandons the sauce, returns to the soup already in the basket, and closes on it from 7 cm away some forty times until the step limit. The arm does not oscillate between the cans; it stalls on the second and switches back to the finished one.

Closing on nothing, and losing what was held



Mean number of *missed closes* per episode: the gripper is commanded shut while no tracked object is within 5 cm. At $K{=}1$ the count is inflated by chatter, since a fresh plan every step can toggle the gripper command many times around one attempt.

Episodes out of 80	$K{=}1$	$K{=}10$	$K{=}25$	$K{=}50$
with at least one missed close	47	43	38	51
with at least one slip	20	21	15	25
slips in total	21	26	18	25

A *slip* is a grasp that made contact and lost it before the gripper was commanded open. Slips happen at a similar rate at every horizon. What differs is what follows: at $K{=}50$ the stale chunk keeps running after the slip and the episode turns into missed closes; at shorter horizons the next call sees the dropped object and tries again.

How failures happen: alphabet soup, start 0, $K{=}50$



step 49



step 70



step 80



step 100

Step	Event	Hand--can	Who decided
0	first call, hand 31 cm from the can	31 cm	call 1
50	second call; the gripper is closed on the can	2.5 cm	call 2 sees the truth
70	can and hand both lifted, still together	4 cm	call 2, open loop
80	can has fallen back to the table; hand keeps carrying	21 cm	call 2, open loop
100	third call; the hand has reached the basket empty	32 cm	call 3 sees the truth
115	gripper closes on nothing; 9 such closes follow	34 cm	call 3 and later

Every call read the real scene. The 32 cm gap at step 100 is not an error of perception at step 50; it is the distance the can fell and the hand travelled inside one 50-step chunk after the grasp slipped, between the second and third still. Clip: demo_alphabet_soup_init00_k50.mp4.

How often a chunk carries on after losing the object

A chunk is counted when, inside one call, the gripper is commanded closed within 5 cm of some object and, later in the same chunk and still commanded closed, that object is at least 15 cm from the hand. Pick-and-place tasks only, 70 starts per horizon.

	$K{=}1$	$K{=}10$	$K{=}25$	$K{=}50$
Episodes with such a chunk	0	1	7	12
...that still succeeded	--	0	1	0

At $K{=}50$ the 12 cases are 12 of 356 chunks. Eleven are on the two cans (soup 5/10, sauce 6/10); none on the milk carton or the bowls. Ten of the same 12 starts succeed at $K{=}25$.

Where in the chunk

The 15 cm separation is crossed at offsets 11 to 45 into the chunk, median 25. The stale plan then runs for 5 to 39 more steps, median 25. A rule that aborted at the crossing would have returned about half of each chunk to the planner.

What this does not explain

The drawer never commands a close at any horizon; its $K{=}50$ losses (4/10 against 8/10) are six runs that drift for all 300 steps. Most table-center losses are also outside this pattern. A gate built on the grasp will fix the cans and leave the drawer alone.

Every five steps, ask one question; replan only when the answer is yes

Base horizon 25. After every five executed actions a rule may empty the queue, so the next step is a new call. The worst case is $K{=}5$; a quiet chunk still runs all 25.

Gate A --- gripper

Have the last 10 commands all said ``close''? If yes, drop the rest of the queue. A closed hand gets a fresh plan every 10 steps; an open hand runs to 25.

Gate B --- fingers

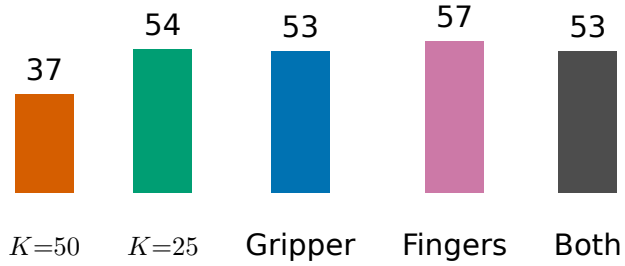
Is the command ``close'' while the fingers are fully shut? Shut fingers with a close command mean nothing is between them. Two such checks in a row drop the queue; a held object keeps the fingers apart.

Both drops the queue when either answers yes. Replayed on the recorded $K{=}50$ traces, Gate A would give a new call about every 18 steps, 13 per episode.

Published alternatives decide at plan time (action entropy, denoising variance, attention entropy) or add a learned visual monitor; both gates here are a few lines of arithmetic on the robot's own state.

Three gates on the same 80 starts, base horizon 25

	$K{=}25$	Gripper	Fingers	Both
Bowl between	**7/10**	**7/10**	6/10	**7/10**
Table center	5/10	**6/10**	5/10	**6/10**
Bowl on stove	6/10	**7/10**	6/10	**7/10**
Alphabet soup	9/10	**10/10**	**10/10**	**10/10**
Tomato sauce	6/10	6/10	**8/10**	6/10
Milk	7/10	8/10	**9/10**	8/10
Soup and sauce	**2/5**	0/5	1/5	0/5
Stove and moka	**4/5**	1/5	**4/5**	1/5
Open drawer	**8/10**	**8/10**	**8/10**	**8/10**
Pooled, 80 starts	54	53	**57**	53
Starts won against $K{=}25$	--	7	6	7
Starts lost against $K{=}25$	--	8	3	8
Network calls per episode	**8.5**	12.0	9.6	12.0



pooled successes out of 80

The finger rule is the one that helps: three starts over the fixed queue for one extra call per episode, with its gains on the three grocery items. The gripper cap wins the same single-object starts and gives back eight, five of them on the two-stage tasks, where cutting every carry forces the policy to re-infer which stage it is in. Combined, the outcomes follow the gripper rule. The drawer is untouched, as both rules predict.

Where the seconds go, and which horizons a real arm could run

The model

`HuggingFaceVLA/smolvla_libero`: about 450 M parameters, 489 weight tensors. Backbone SmolVLM2-500M-Instruct; the action expert reuses its first 16 layers at 0.75 width and takes 10 Euler steps per chunk.

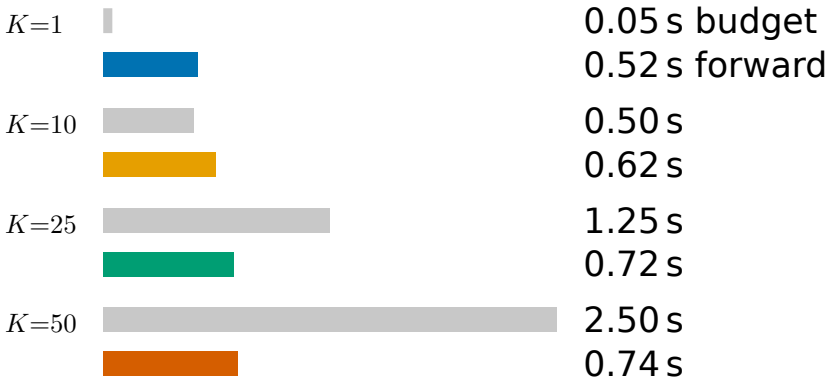
Measured on an RTX 3060 Laptop (6 GB)

	$K{=}1$	$K{=}10$	$K{=}25$	$K{=}50$
One forward	0.52 s	0.62 s	0.72 s	0.74 s
Network, episode	111 s	13 s	6.1 s	3.7 s
Wall clock, episode	148 s	49 s	42 s	44 s
Arm motion, 20 Hz	10.7 s	10.4 s	10.1 s	11.6 s

The simulator is the other 35--40 s. A physical arm would spend only the last two rows.

Does one forward fit inside the motion it commands?

Grey: the time budget $K/20$ seconds. Colour: the measured forward.



$K{=}1$ and $K{=}10$ overrun their budget, so a synchronous controller would stall between chunks. $K{=}25$ is the first horizon where the forward fits, and the adaptive gates keep its worst case at 5 steps (0.25 s), which would need the call to overlap with motion. $K{=}50$ fits with room to spare and pays for it in open-loop losses.